Chapter 28 | Written solutions

Learning from Tables

All eleven exercises connect feature availability, fixed row encodings, feature tokens, context weighting, and honest evaluation. Exercises 1-4 use hand calculations, 5-8 use the canonical NumPy functions, and 9-11 design comparisons for deployment. The numerical fixtures illustrate concepts; they are not measured model benchmarks.

The written PDF contains the complete explanations, tables and three plots. This notebook and its PDF add executable checks and recorded results.

1. Decide when each feature exists

Suppose a customer books a Friday appointment on Wednesday. The channel and scheduled date entered on Wednesday are available, even though Friday is in the future. A subsequent rescheduling is a different version of the date. The cancellation fee actually charged is determined later and would leak the outcome.

Earlier completed bookings are available only if their completion had already been recorded and was retrievable on Wednesday. A reservation made earlier for Thursday is still pending at prediction time. Event time and information arrival time both matter: a completed booking whose record arrives late is also unavailable. The miniature timeline checks this distinction.

2. Separate new and missing categories

Keep the columns in the order: filled value, number missing, mail, web, category missing, category unknown. Fill only a missing number with 5.

Input row	Encoded row	Reason
missing, web	$(5, 1, 0, 1, 0, 0)$	The number is filled; web is known.
2, missing	$(2, 0, 0, 0, 1, 0)$	The number is observed; the category is absent.
2, phone	$(2, 0, 0, 0, 0, 1)$	Phone is present but outside the vocabulary.

The two final flags carry different information. An observed 5 would have numeric-missing flag 0; a filled-in 5 has flag 1. There are $K + 4 = 6$ columns.

3. Compute a context-weighted prediction

Multiply each label by its fixed weight and add:

$$\widehat{p} = 0.2(1) + 0.3(0) + 0.5(1) = 0.7.$$

Changing the second label from 0 to 1 adds exactly its weight, 0.3, so the new estimate is 1. All labels are now positive. This experiment holds weights fixed. Changing a similarity score would instead renormalize every weight, as Exercise 7 demonstrates. Neither experiment by itself implements TabPFN.

4. Distinguish probabilities from decisions

At threshold 0.5, both predictors decide $(0, 1)$. With labels $(0, 1)$ both decisions are correct, but their probability errors differ:

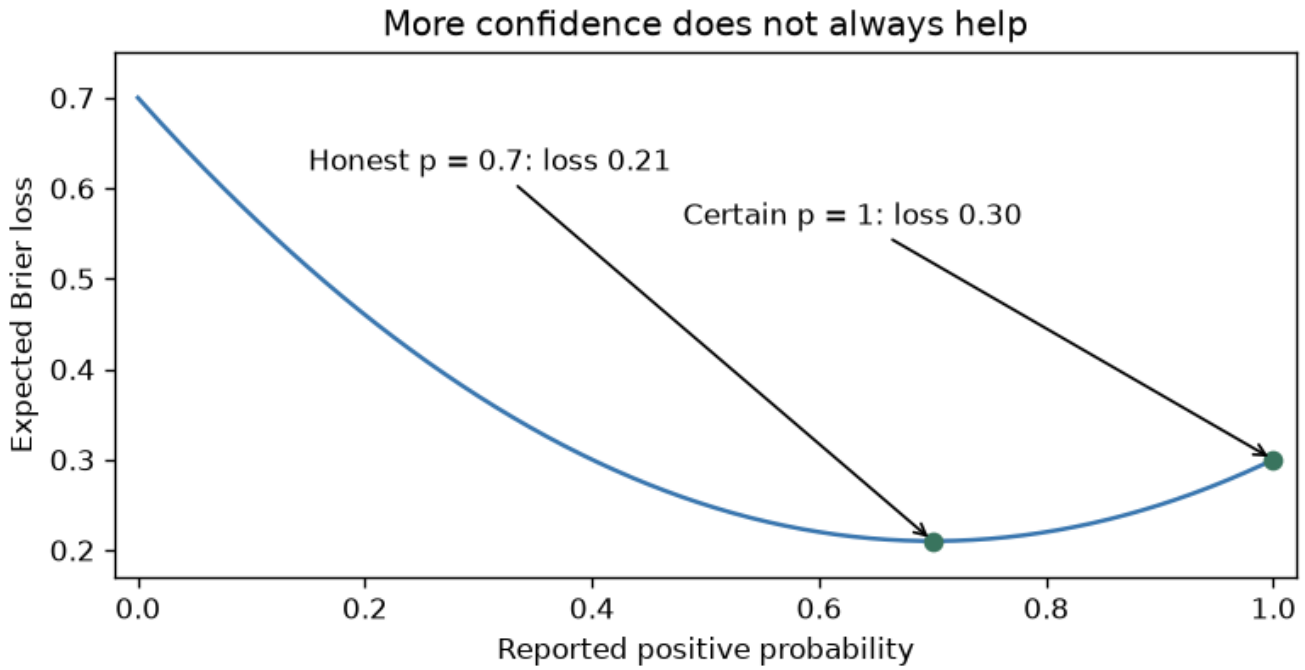
$$B_A = \frac{0.2^2 + (0.7 - 1)^2}{2} = 0.065, \qquad B_B = \frac{0.4^2 + (0.6 - 1)^2}{2} = 0.16.$$

Thresholding discards the distance between a probability and an outcome. If the labels are instead $(1, 0)$, both predictors get both decisions wrong:

$$B_A = \frac{(0.2 - 1)^2 + 0.7^2}{2} = 0.565, \qquad B_B = \frac{(0.4 - 1)^2 + 0.6^2}{2} = 0.36.$$

A is more confident but now receives the worse score. Confidence helps when it is justified, and hurts when it points in the wrong direction. These two cases cannot establish calibration on a population.

The expected score makes the same point before an outcome is known. If the true positive probability is $\theta = 0.7$, reporting p incurs $\theta(1 - \theta) + (p - \theta)^2$. Its minimum is 0.21 at $p = 0.7$. Reporting certainty $p = 1$ raises the expected loss to 0.30. The curve below is calculated from the two possible outcomes, not fitted to data.



5. Keep a held-out row out of fitting

A blank in the days column needs a replacement value. The encoder learns that value and the known channel names from the four training orders. Each position in `train_numeric` and `train_channels` refers to the same order. `None` marks a blank.

The observed training days are 2, 4, and 8. Their middle value is 4, so the fitted median is 4. The distinct observed channels in alphabetical order are `shop` and `web`. That order sets the positions for the two known-channel indicators. Each indicator is 1 when its condition holds and 0 otherwise.

Apply this fit to the three held-out orders. The row with a blank days value gets 4 and a 1 in `days missing`. Its channel `kiosk` was unseen during training, so `channel unknown` is 1. The row with 6 days and channel `shop` keeps 6 and activates `shop`. The row with 100 days and a blank channel keeps 100 and activates `channel missing`. Both `kiosk` and the blank channel leave the `shop` and `web` entries at 0.

Append the order with 12 days and channel `mobile`. Pass the same median and levels to `encode_rows`. The extra row keeps its observed value 12 and uses the existing `channel unknown` flag. The four-row result still has six columns, and its first three rows match the original output. The held-out values 100 and 12 appear in their own rows. The median used to fill the blank came from the training days $(2, 4, 8)$.

6. Test the effect of an unseen category

Use the fit from Exercise 5 and hold the days value at 6. Change only the channel. The two numerical entries should be $(6, 0)$ in every row because 6 is an observed number. The last four columns tell us which of the three channel cases applies.

For `shop`, the 1 is in column 3, the known `shop` indicator. For `kiosk`, it moves to column 6 because the channel is present and absent from the training vocabulary. For `None`, it moves to column 5 because the channel itself is missing. Column 4, the indicator for `web`, is 0 throughout. The numerical columns agree exactly. The distinction between an unseen channel and a blank one is carried by two separate categorical flags.

The unknown flag retains the fact that a supplied category was unseen. It does not retain its name: mobile and kiosk have identical encodings when the numerical value is held fixed. A model trained with an always-zero unknown column has no observed outcomes for that indicator. A fallback or a training strategy for rare and unseen levels must be designed and validated separately.

7. Trace the influence of a context row

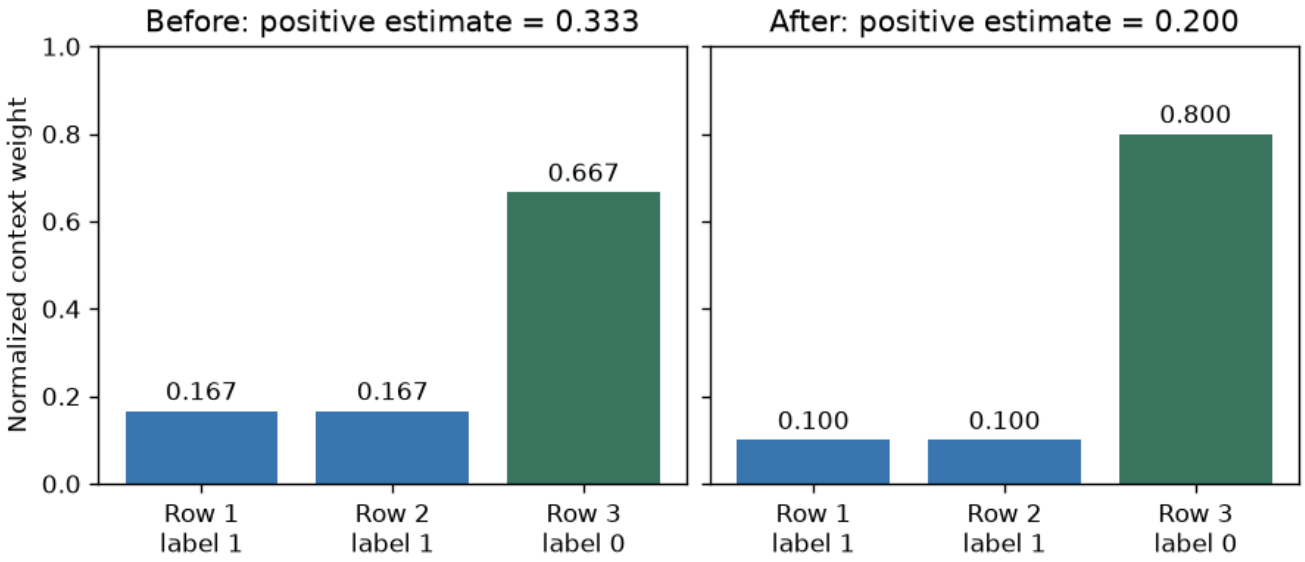
Think of a query order whose label we want to predict. The first two context orders have positive labels. The third has a negative label, and the query receives a higher similarity score for that order. We will raise the third score and see how much influence it gains.

Softmax exponentiates the scores and divides by the sum of those exponentials. With scores $(0, 0, \ln 4)$, the exponentials are $(1, 1, 4)$, which sum to 6. The weights are $(1/6, 1/6, 4/6)$. The toy predictor multiplies each weight by its row's label and adds the products. Only the first two labels are 1, so the initial positive-class estimate is

$$\widehat{p} = \frac{1}{6}(1) + \frac{1}{6}(1) + \frac{4}{6}(0) = \frac{1}{3}.$$

Raising the last score by $\ln 2$ makes it $\ln 8$. Its exponential doubles from 4 to 8, while the first two stay at 1. The new denominator is 10, so the weights are $(1/10, 1/10, 8/10)$. The estimate becomes $1/10 + 1/10 = 1/5$. The table below checks both predictions and all six weights against these hand calculations.

The negative row takes $8/10$ of the weight after the change, compared with $4/6$ before. Because the three weights must sum to one, less weight remains for the two positive rows. The negative row's extra weight contributes zero to the weighted label sum, and the estimate drops from about 0.33333 to 0.20000. A larger similarity means more influence for that context row. The direction of the probability change depends on its label.



8. Score two confidence levels

Suppose two predictors classify the same four orders. Each gives a probability of label 1. Let A give $(0.2, 0.7, 0.8, 0.1)$ and B give $(0.45, 0.55, 0.6, 0.4)$. The observed labels are $(0, 1, 1, 0)$.

At threshold 0.5, assign label 1 when $p_i \geq 0.5$ and label 0 otherwise. Each predictor gives $(0, 1, 1, 0)$, so both get all four decisions right and achieve accuracy $4/4 = 1$.

In the chapter's binary convention, each case contributes the squared difference between its positive-class probability p_i and its observed label y_i. The Brier score is the mean of those contributions.

$$\text{Brier} = \frac{1}{4}\sum_{i=1}^{4}(p_i - y_i)^2.$$

Case	Label	A probability	A squared error	B probability	B squared error
1	0	0.20	0.0400	0.45	0.2025
2	1	0.70	0.0900	0.55	0.2025
3	1	0.80	0.0400	0.60	0.1600
4	0	0.10	0.0100	0.40	0.1600

For example, the second case has label 1. A assigns it 0.7, giving $(0.7 - 1)^2 = 0.09$. B assigns it 0.55, giving $(0.55 - 1)^2 = 0.2025$. The sums are 0.18 for A and 0.725 for B, so the scores are 0.045 and 0.18125 respectively.

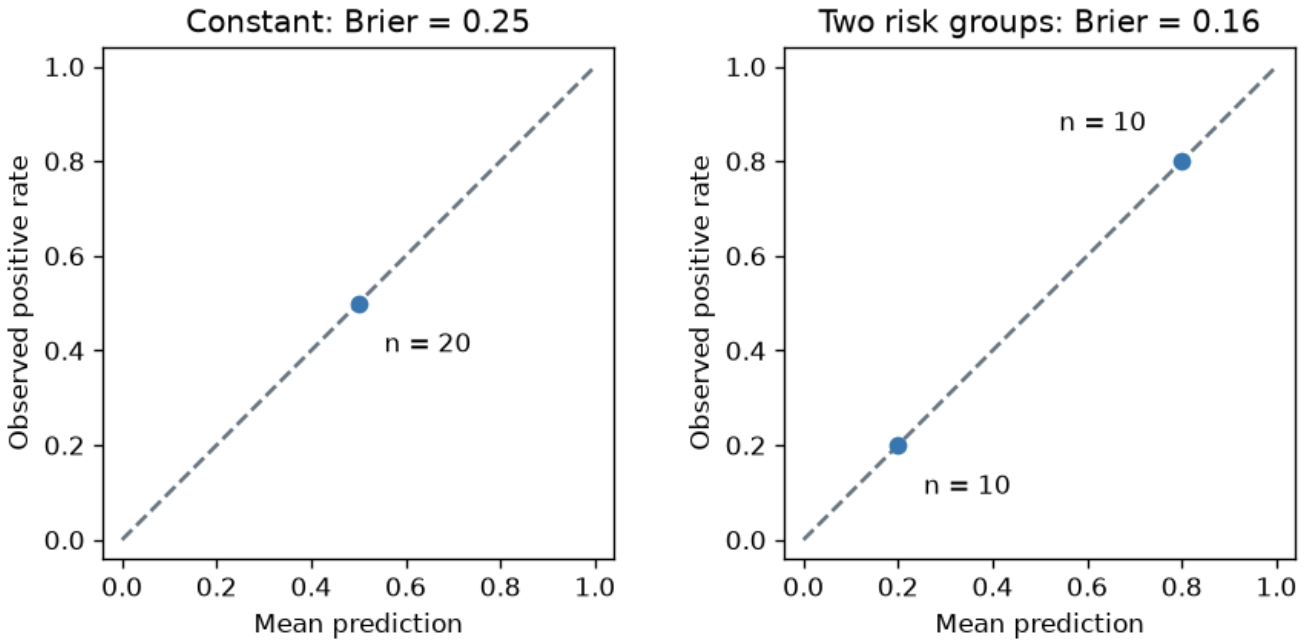
A and B make the same decisions, but A assigns probabilities closer to the observed labels in every case. A therefore gets the lower Brier score on these four cases. Accuracy at 0.5 gives both predictors the same credit because it only sees the decisions.

Calibration asks whether events assigned a given probability occur about that often. For example, among many independent orders that receive predictions near 0.7, the observed fraction of positive labels should be near 0.7. A single order receiving 0.7 has only one outcome, so it tells us little about that frequency.

To compare calibration, collect many held-out predictions and outcomes from the intended deployment population. Group each predictor's probabilities into the same ranges and compare the mean predicted probability with the observed positive rate in each range. Report how many orders fall in each range and attach uncertainty intervals, accounting for repeated customers if their orders are related. This can reveal where a predictor's probabilities run systematically high or low. The Brier scores here compare overall probability error on the four given cases.

Now construct the separate 20-case calibration example. The first ten labels contain two positives; the next ten contain eight. A constant predictor assigns 0.5 to all cases, matching the overall rate $10/20$. A second predictor assigns 0.2 to the first group and 0.8 to the second, matching $2/10$ and $8/10$. Thus both match observed frequencies at their prediction values.

The constant predictor has loss $(0.5 - y)^2 = 0.25$ on every case. In the first group, the second predictor's mean loss is $(2(0.8)^2 + 8(0.2)^2)/10 = 0.16$; the second group gives the same result. It distinguishes risk groups and has lower Brier loss despite both predictors' perfect empirical calibration. This deliberately constructed example is not evidence that 20 observations suffice to validate calibration in deployment.



9. Split for the intended customer

For new customers, one valid design assigns 350 customer IDs to training, 75 to validation, and 75 to test. Every order follows its customer's split. Verify disjoint ID sets and report row and class counts: customers with five orders contribute more rows than customers with one. Fit preprocessing and models on training customers and choose settings on validation customers. For a first-order deployment question, score the first eligible order only. If later orders from held-out customers are included, state that population or report them separately. History features must still obey each order's timestamp.

For later orders from existing customers, use global time cutoffs. For example, train on orders placed through April, validate on June, and test on August. An August test customer must have an order known before August; apply the analogous rule to June. Earlier and later rows from the same ID are allowed because that is the intended task. Separate per-customer cutoffs could wrongly train on one person's August order while predicting another person's June order.

If labels require 30 days to mature, leave May and July for outcomes to arrive. An April 30 order matures on May 30, before June evaluation; a June 30 label matures on July 30, before August testing. Recent orders may supply history while their labels remain unavailable for fitting or context. Freeze each fit and selection using only outcomes then known; score August outcomes after their observation windows finish. A customer with only an earlier order has no later test order. Report eligible customers and rows.

The fixture gives customer i exactly $1 + (i \bmod 5)$ orders. It checks split counts, not predictive performance. Averaging order losses weights frequent customers more; averaging each customer's mean loss gives equal customer weight.

10. Compare three methods fairly

Compare logistic regression, a gradient-boosted tree ensemble and a frozen TabPFN classification checkpoint on one fixed task. Define the label and prediction time, then give every method the same available input fields. Exclude outcome-revealing fields such as later refund amounts.

Use a deployment-matched split, such as the grouped new-customer design. Draw nested subsets of 25, 50, 100 and 200 training customers. Use every eligible labeled order from each chosen customer and give the same labeled rows to all three methods. Each larger subset contains the smaller one. Report the actual row counts, and use sizes supported by the chosen checkpoint. Repeat with recorded seeds. A prespecified sampling rule based only on training labels can require both classes in the smallest subset; logistic classification needs both. Disclose that rule because it changes the sampling design.

At each size, fit the logistic weights, tree model and data-dependent preprocessing from that subset only. Different methods may have different appropriate preprocessing. The frozen model receives the subset's features and labels as context; queries supply available features without true labels. Record the checkpoint revision, feature types and context construction. Equal task-label budgets do not erase the pretrained model's earlier synthetic training.

Use validation to select regularization, tree settings, context choices and decision thresholds, with disclosed tuning budgets. Keep test cases fixed at every size and never use their labels for choices. Report decision and probability scores, including Brier loss, with variability across subset draws. Account for customer dependence when estimating uncertainty. Separate fit, checkpoint load, context preparation, first-query and repeated-query costs; report hardware and memory. A changing learning curve reflects both the amount and composition of the selected training information.

The executable fixture below checks nested subsets and row budgets. Its labels are synthetic and it trains no models; producing invented benchmark curves would not answer this design exercise.

11. Investigate unreliable probabilities

Suppose the predictions in the bin $[0.7, 0.8)$ average 0.74 and its observed positive rate is 0.50. Compare 0.50 with the actual mean 0.74, not the midpoint 0.75: the gap is -0.24. First count orders, positives and distinct customers. Two positives in four cases and 200 in 400 give the same rate with very different precision. Confirm that all outcome windows have matured; pending returns can make an apparent low rate misleading.

Measure overlap with training and earlier validation customers. Report new and existing customers separately when counts permit. Repeated orders reduce the amount of independent information. Resample whole customers for a customer-based uncertainty analysis and inspect time blocks if common shocks create temporal dependence. A pooled interval can hide a change concentrated in one week.

Compare inputs as they existed at prediction time: channel shares, missing and unseen-category rates, numerical distributions and the fraction of new customers. Compare positive rates after labels mature. Feature drift concerns the inputs; outcomes can also change at similar inputs. Evaluate the same frozen model with the same bin boundaries in an earlier validation period, reporting mean probabilities, observed rates and counts.

Treat the latest data examined above as a diagnostic period. Fit any recalibration on a separate labeled period, for example a logistic map of the model's log-odds $\ln(p/(1-p))$, with a fixed endpoint-clipping rule. Freeze the map and evaluate original and adjusted probabilities on the same later untouched test orders. Compare reliability, counts and Brier loss, with uncertainty. Do not replace all future predictions by the observed diagnostic rate and call that validation.

The code checks half-open bin boundaries and the denominator. It intentionally does not fit a recalibrator to the diagnostic outcomes.